The Extremal Principle — problem 3.42

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It is known that the numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ are both permutations of $1, 1/2, \ldots, 1/n$. In addition, we know that $a_1 + b_1 \geq a_2 + b_2 \geq \cdots \geq a_n + b_n$. Prove that $a_m + b_m \leq 4/m$ for all m from 1 to n.